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A Unique Perspective on Vulnerability and Route Dynamics in the Global Liner Shipping Network

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Abstract

The resilience of the Global Liner Shipping Network (GLSN) has been examined in various studies under different scenarios, with some indicating resilience to random attacks but vulnerable to deliberate attacks, particularly targeting high-degree nodes. This study uniquely focuses on assessing GLSN resilience as adaptability using two different scenarios of real-world data in the face of a significant disruption like pandemic. Through comprehensive network analysis, which includes route dynamic assessment in two different scenarios, we observe that the GLSN demonstrates moderate adaptability to substantial disruptions. Our investigation extends to the dynamic evolution of shipping routes during disruptive periods. Our findings reveal that top-degree nodes are vulnerable for unique routes, while both the highest initial and end ports are vulnerable for similar cross-year routes. Additionally, all scenarios have various impacts on both the adaptability index and traffic change across the years.

Keywords Global liner shipping network · Pandemic · Resilience · Vulnerability · Route

1 Introduction

In general, resilience is defined as the capacity to adapt to new challenges or restore to a normal state after disruption [1]. Resilience assessment involves considerations of simulation, disruption, and measurement. Simulation entails determining the scenarios used to assess resilience, while disruption involves specifying the type of

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attack or removal strategy employed to closely mimic real-world events. Measurement is employed to evaluate and interpret the results of the simulation. Ultimately, it will be determined whether the network exhibits resilience or not. Various strategies, such as random removal [2], deliberate attacks targeting high-degree nodes [3], and multi-centrality assessments [4], have been utilized. Additionally, attacking strategies may differ, including static [5] and dynamic states [4], as well as partial removal [6].

Resilience assessments frequently narrow their focus to specific scenarios, such as the impact of natural disasters like the 2011 tsunami on Kobe Port [3] or events similar to those experienced by Shanghai Port in 2020 [4, 7]. However, when evaluating the resilience of the global liner shipping network, it is crucial to acknowledge that nodes and links primarily represent the sequence of routes taken by ships rather than discrete start and end points. Therefore, assessing route dynamics emerges as the most suitable method for evaluating the network's resilience, specifically resilience as adaptability [8].

To this end, two distinct scenarios are proposed for assessment: the "as is" scenario and the deliberate node removal scenario. The "as is" scenario involves analyzing existing data, with 2019 data serving as the baseline, 2020 representing a disruption period, and 2021 indicating a rebounding period. Conversely, the deliberate node removal scenario comprises three scenarios: the group of highest centrality node removal, the group of most initial and end port removal.

For the assessment of resilience as adaptability, we will use Jaccard [9] similarity analysis where the adaptability index will be generated as a measure for the network adaptability. Additionally, it is worth noting that existing simulations often target high-degree nodes for removal [4, 10], under the assumption that they are more susceptible to disruption.

When assessing node vulnerability, we rely on a range of measures, encompassing centrality degree, most initial port point, and most end point port. A centrality degree offers valuable insights into port activity by considering both incoming and outgoing traffic, effectively pinpointing the busiest ports within the network. The designation of the most initial port point typically denotes where ships commence their journeys, while the identification of the most end point port signifies the termination point of these voyages. Our study aims to assess the Global Liner Shipping Network's response to the recent pandemic disruption using real-world data. We take a unique approach to vulnerability assessment, aiming to identify nodes most susceptible to disruption.

The structure of the remaining paper will unfold as follows: Sect. 2 will present the literature review. Section 3 will cover data and methodology. Section 4 will delve into results and discussions. Section 5 will conclude the paper.

2 Literature Review

The resilience of the Global Liner Shipping Network (GLSN) has been a significant area of research, with numerous studies contributing to our understanding of its vulnerabilities and adaptive capacities. Xu et al. (2022) investigate the GLSN's

vulnerability, particularly focusing on cascading failures during interruptions such as the COVID-19 pandemic. By employing a bipartite network model, they reveal the dynamic nature of GLSN's vulnerability and emphasize the importance of adjusting port rotations. Their recommendations to skip failed ports rather than choosing alternatives and to expand port capacity for long-term robustness underscore critical strategies for maintaining network stability [6]. Our research builds on this by examining how adaptive routing algorithms can further enhance these resilience strategies.

In the realm of adaptability, Silva et al. (2017) introduce "k-adaptive," a novel routing approach designed to handle uncertainty in network loading. This method allows decision-makers to iteratively refine multiple potential solutions, thereby improving overall network quality [11]. Similarly, Nemeth (2015) underscores the importance of adaptive routing algorithms in managing Internet traffic, proposing a rate-adaptive multipath routing model with new centralized algorithms and a hybrid distributed-centralized approach [12]. Sharkey et al. (2021) further highlight the significance of network optimization methods in enhancing adaptability to future uncertainties, calling for extensive research to maximize network adaptability [8]. Our study extends these findings by applying adaptive routing principles specifically to GLSN, demonstrating their potential to mitigate disruptions and improve resilience.

Bai et al. (2023) contribute a two-dimensional framework for analyzing GLSN's static and dynamic resilience. Using vessel movement and AIS data, they apply the clique percolation method and a knock-on effect simulation model to uncover the influence of overlapping structures and traffic flow on GLSN's static resilience, particularly to random disruptions [4]. Their dynamic analysis provides valuable insights into disruption propagation, aiding contingency planning. Our work integrates these insights with adaptive routing techniques to enhance both the static and dynamic resilience of GLSN.

Calatayud et al. (2017) and Liu et al. (2018) both utilize complex network analyses to assess vulnerability in international freight flows and maritime supply chains, respectively. Calatayud et al. advocate for multiplex network modelling to manage supply chain risks, highlighting the need to consider a country's position in the maritime network [3]. Liu et al. emphasize the robustness of maritime networks to random failures but increased vulnerability to targeted attacks on hub ports, stressing a comprehensive approach to identifying weaknesses [2]. Building on these studies, our research explores how adaptive routing and strategic port rotation can address these vulnerabilities and improve overall network resilience.

Dui et al. (2021) address Maritime Transportation Systems (MTS) resilience through a dynamic optimization model that prioritizes the recovery of critical supply nodes and connecting links, considering socio-economic impacts [13]. Qin et al. (2023) further investigate port node resilience using a three-dimensional model, analyzing data from China's container ports to identify trends and dominant factors influencing resilience [14]. Our study complements these approaches by incorporating adaptive routing strategies to enhance the resilience of both MTS and port nodes within GLSN.

Narasimha et al. (2021) and Notteboom et al. (2021) analyze the impact of COVID-19 on seaport transportation and the shipping industry. Narasimha et al. reveal significant negative growth in cargo and vessel traffic in India, stressing the industry's lack of preparedness and the need for resilient recovery strategies [15]. Notteboom et al. compare the impacts of COVID-19 and the 2008–2009 financial crisis on container ports, highlighting differences in adaptation mechanisms and operational adaptability [16]. Our research aims to integrate these findings with adaptive routing to develop more robust recovery strategies for GLSN in the face of global disruptions.

Camur et al. (2024) provide an in-depth exploration of centrality metrics in network analysis and optimization, particularly group centrality. They highlight the importance of identifying influential groups of nodes with connectivity constraints and call for further research into probabilistic group centrality metrics and stochastic optimization techniques [17]. Our work leverages these centrality metrics to enhance the identification and protection of critical nodes within GLSN, contributing to a more resilient network structure.

In summary, while existing studies have extensively explored GLSN resilience through various models and frameworks, our research uniquely integrates adaptive routing algorithms and network optimization techniques to address both static and dynamic resilience. By building on previous findings and introducing novel approaches, we aim to enhance the overall robustness of GLSN and provide practical recommendations for improving its adaptability to future disruptions.

3 Data and Methodology

3.1 Data

We utilized data from alphaliner.com for the years 2019, 2020, and 2021 to gather information on global liner shipping routes [18]. The dataset includes crucial details such as service ID, carrier, capacity, frequency, duration, and rotation.

Once the data was obtained, we performed a thorough cleaning process to ensure its accuracy and reliability. This involved addressing issues such as misspelled entries, redundant information, empty strings, and numerical inconsistencies. By carefully cleaning the data, we were able to enhance its quality and eliminate any potential errors.

The resulting dataset consists of 2601 ports located around the world, along with approximately 29903 link that span across various regions globally. This comprehensive coverage enables a robust analysis of the global liner shipping network, facilitating a deeper understanding of its structure and dynamics.

3.2 Methodology

We adopted the L-space approach [19] to construct directed graphs representing the global liner shipping network for the years 2019, 2020, and 2021. Subsequent to the construction of these networks, our analytical focus shifted towards visualizing their structures. Given the sparsity of the networks, our topological analysis was tailored

to key metrics, including the count of nodes, edges, total shipped weight, average clustering coefficient, density, and the size of the strongly connected components.

$$C = \frac{\text{Number of directed triangles}}{\text{Number of connected triples}} \quad (1)$$

where C stands for the clustering coefficient, *Number of directed triangles* is the number of sets of three nodes where each node has a directed edge to another node in the set, forming a cycle, and *Number of connected triples* is the number of sets of three nodes connected by either directed or undirected paths.

$$\text{Density} = \frac{|E|}{|V| * (|V| - 1)} \quad (2)$$

where $|E|$ is the total number of directed edges in the graph, and $|V|$ is the total number of vertices (nodes) in the graph.

A strongly connected component (SCC) in a directed graph is a maximal subgraph in which every pair of vertices (nodes) is mutually reachable. This means that for any two nodes u and v within the SCC, there exists a directed path from u to v and a directed path from v to u . In other words, all nodes in an SCC are interconnected such that travel between any two nodes within the component is possible in both directions.

Formally, for a directed graph $G = (V, E)$, where V is a set of nodes (ports) and E is a set of directed edges between nodes. An SCC is defined as a subgraph $G' = (V', E')$ such that for every pair of node $u, v \in V'$, there exists a directed path from u to v and from v to u .

Moving forward, we delved into route analysis, aiming to quantify the number of operational routes in each respective year. Additionally, we conducted a comparative assessment of route similarities within the same year and across different years. This analysis allowed us to discern the relative busyness of specific routes. Here is how it is done [9]:

$$J(R_i, R_j) = \frac{|N_i \cap N_j|}{|N_i \cup N_j|} \quad (3)$$

where $J(R_i, R_j)$ is the similarity index, N_i is the set of nodes in route R_i , and N_j is the set of nodes in route R_j . To consider the two routes similar, we use the following mathematical condition:

$$\text{Similar}(R_i, R_j) = f(x) = \begin{cases} 1, & \text{if } J(R_i, R_j) > 0.9 \\ 0, & \text{otherwise} \end{cases} \quad (4)$$

$$\text{Similar Routes} = \{ (R_i, R_j) | \text{Similar}(R_i, R_j) = 1 \} \quad (5)$$

Continuing our exploration, we turned our attention to develop the adaptability index. To understand how routes in the Global Liner Shipping Network (GLSN) adapt to changes over time, we developed an adaptability index by comparing traffic

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changes to traffic volumes across different years. The adaptability index provides a measure of how well the network adapts to disruptions or shifts in demand.

1. Traffic volume (TV) represents the amount of cargo travelling on a specific route within a given year. For each route R_i in the GLSN, the traffic volume $TV_{i,y}$ for year y is recorded.
2. Traffic change (TC) is defined as the difference in traffic volume for each route between two consecutive years. For route R_i , the traffic change from year y to year $y + 1$ is calculated as:

$$TC_{i,y,y+1} = TV_{i,y+1} - TV_{i,y} \quad (6)$$

This value can be positive, indicating an increase in traffic, or negative, indicating a decrease.

3. The adaptability index (AI) for each route is a measure that reflects how adaptable the route is to changes in traffic volume. It is calculated by normalizing the traffic change by the average traffic volume over the two years. For route R_i , the adaptability index from year y to year $y + 1$ is given by:

$$AI_{i,y,y+1} = \frac{TC_{i,y,y+1}}{\left(\frac{TV_{i,y} + TV_{i,y+1}}{2}\right)} \quad (7)$$

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This normalization helps in understanding the relative change in traffic, accounting for the scale of the traffic volumes.

4. To get an overall sense of the network's adaptability, we calculate the average adaptability index and average traffic change across all routes for each year pair. These aggregate measures provide insights into the general trends in network adaptability and traffic fluctuations.

$$ATC_{y,y+1} = \frac{1}{N} \sum_{i=1}^N TC_{i,y,y+1} \quad (8)$$

where ATC is the average traffic change, and N is the total number of routes.

$$AAI_{y,y+1} = \frac{1}{N} \sum_{i=1}^N AI_{i,y,y+1} \quad (9)$$

where AAI is the average adaptability index, and N is the total number of routes.

By using these calculations, we can analyze how the GLSN adapts over time. For instance, a high positive adaptability index indicates that the network routes can effectively handle increased traffic, while a negative index points to challenges in accommodating traffic changes.

Moving forward to vulnerability assessment, to comprehensively assess network vulnerability, we adopt a systematic approach using the routing adaptability index and traffic change as our metrics. These indices help us measure the network's adaptability to changes. We employ three deliberate removal strategies: targeting the

top 20 highest degree ports, the top 20 most initial ports, and the top 20 most end ports. This allows us to identify which routes are most adaptable and which are most vulnerable to such disruptions. Furthermore, we analyze which strategy is most detrimental to specific routes, aiding stakeholders in mitigating disruption impacts on route adaptability. Our study consists of two scenarios: the “as is” scenario, where we analyze the dataset to understand the network’s current state, and the deliberate attack scenario, where we implement various removal strategies to assess both route adaptability and vulnerability under different attack scenarios.

4 Result

4.1 As is Analysis

The initial phase involves the meticulous construction of the Global Liner Shipping Network (GLSN) for the years 2019, 2020, and 2021. The subsequent visualization of these networks provides a comprehensive understanding of their structures. Figures 1, 2, and 3 visually encapsulate the essence of these networks.

Figures 1, 2, and 3 offer a visual insight into the dynamics of GLSN across the considered years. Notably, GLSN in 2020 (Fig. 2) appears sparser in comparison to 2019 (Fig. 1) and 2021 (Fig. 3). This observation serves as a compelling piece

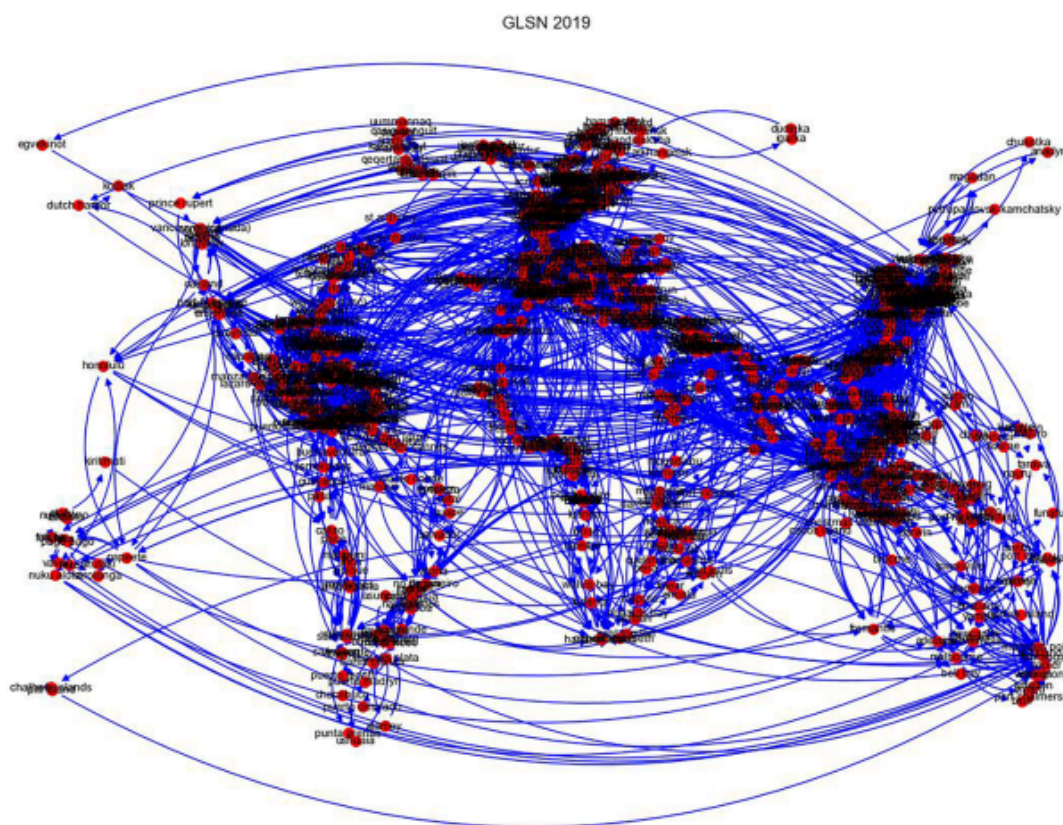


Fig. 1 Visual of Global Liner Shipping Network (GLSN) in 2019. The red dots are ports and the blue lines are the routes



Fig. 2 Visual of Global Liner Shipping Network (GLSN) in 2020. The red dots are ports and the green lines are the routes

of evidence, indicating a significant disruption [4, 7] in the shipping network during the year 2020.

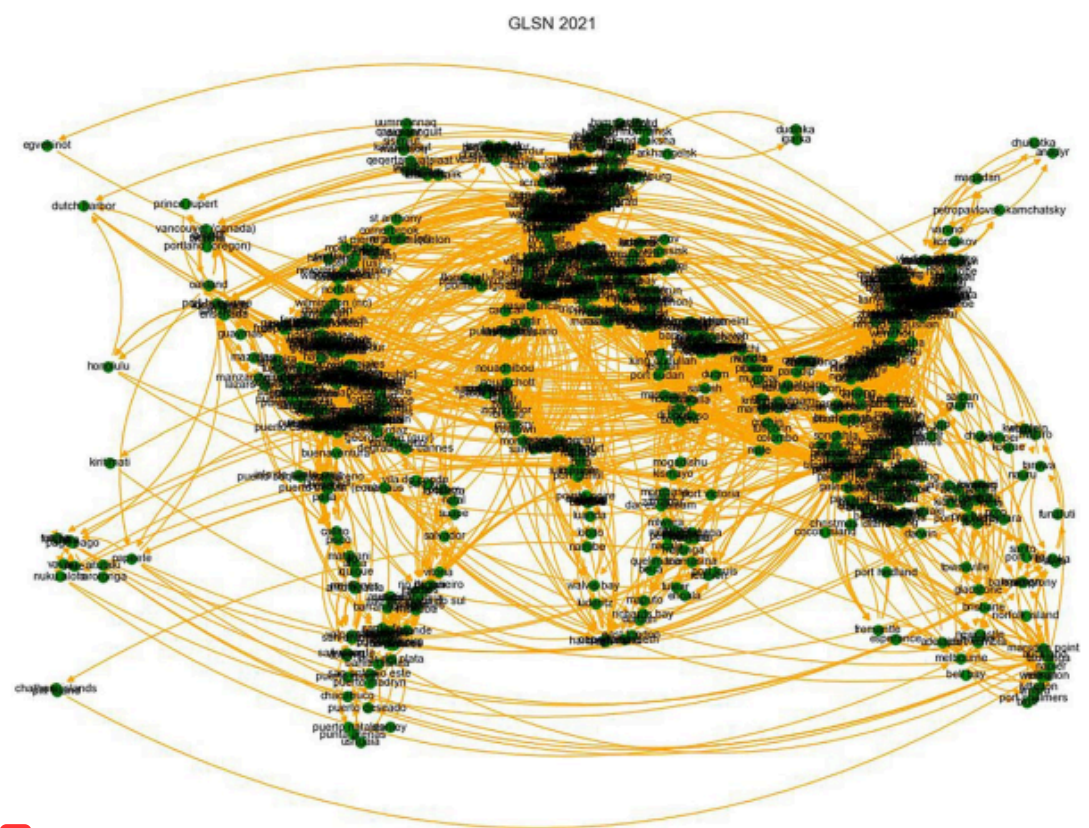
Despite the apparent disruption in 2020, the resilience of the network becomes evident as it rebounds in 2021. The visualization portrays a renewed and robust network structure, signifying the adaptability and recovery of the global liner shipping network.

This visual exploration lays the foundation for subsequent analyses, allowing for a more in-depth investigation into the network’s properties, routes, and vulnerabilities during these critical years.

Table 1 offers a comprehensive overview of key properties within the Global Liner Shipping Network (GLSN) across the years 2019, 2020, and 2021 (Table 1). Notably, a consistent trend emerges, revealing a decline in ports, links, and weights in 2020 followed by a substantial rebound in 2021. This distinctive pattern signifies a temporary disruption in 2020 [7], succeeded by a resilient recovery that surpasses previous network levels in 2021.

4.1.1 Ports, Links, and Weights

A clear trend showcases a decline in these fundamental metrics in 2020, reflecting the disruptive impact on the network during that period. The subsequent surge in 2021 indicates a robust recovery, showcasing the system’s resilience.



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Fig. 3 Visual of Global Liner Shipping Network (GLSN) in 2021. The green dots are ports and the yellow lines are the routes

Table 1 Global liner shipping network basic property comparison

	2019	2020	2021
Number of nodes	936	730	982
Number of edges	5093	3979	5265
Total weight shipped	279,687,870	263,393,021	299,488,907
Average clustering coefficient ¹	0.259	0.319	0.270
Density ²	0.0058	0.0074	0.0054
Strongly connected components	54	4	54

¹Calculated using Eq. 1

²Calculated using Eq. 2

4.1.2 Average Clustering Coefficient and Density

The noticeable increase in the average clustering coefficient in 2020 corresponds with the observed decrease in the number of nodes, edges, and weight in GLSN during that year. This suggests a strategic response, with nodes forming clusters or interconnected groups, potentially adapting to disruptions.

4.1.3 Shifts in Structure

The decrease in both the average clustering coefficient and density in 2021 signals a shift in the network structure. Reduced density implies a reconfiguration or streamlining of shipping routes, while the decline in the clustering coefficient indicates fewer tightly-knit clusters within the network.

4.1.4 Strongly Connected Components

Dynamic shifts in the number of strongly connected components further illustrate network adaptations. The substantial reduction from 54 to 4 components in 2020 mirrors disruptions, leading to a reconfiguration. The subsequent rebound to 54 components in 2021 signifies a robust recovery and restructuring, highlighting the network’s resilience in the face of external pressures.

This comprehensive analysis provides a nuanced understanding of the GLSN’s evolution, offering valuable insights into its response to disruptions and its subsequent recovery and adaptability [1, 2, 16].

By utilizing Eqs. 3–5, Table 2 provides a comprehensive analysis of the dynamic routes within the Global Liner Shipping Network (GLSN) for the years 2019, 2020, and 2021 (Table 2). The shipping industry, marked by its adaptability, crafts unique networks of routes to efficiently transport cargo, and understanding the evolution of these routes reveals insightful trends.

4.1.5 Route Evolution

2019 Infrastructure: In 2019, the GLSN exhibited a robust infrastructure, boasting a total of 1582 distinct routes, highlighting the expansive nature of the network [20, 21].

Impact of 2020 disruptions: The subsequent year, 2020, witnessed a notable contraction, with the number of routes decreasing to 1117. This dip can be attributed to the disruptions, likely a consequence of the global pandemic, reflecting the industry’s response to unprecedented challenges [22].

4.1.6 Similar Routes Analysis

Instances of similar routes: Examining the concept of “similar routes”—services covering identical sequences of ports—reveals insightful trends. In 2019, there were 99 instances of similar routes, suggesting multiple services catering to the same destinations. This count diminished to 22 in 2020, indicative of challenges faced during

Table 2 Analysis of route dynamics in the Global Liner Shipping Network (GLSN) 2019, 2020, 2021

	Number of similar and unique routes				Total routes
	2019	2020	2021	Unique	
2019	99	300	703	480	1582
2020	300	22	409	386	1117
2021	703	409	84	422	1618

the disruptive period. Encouragingly, the situation improved in 2021, with 84 similar routes identified, marking a recovery in services and a return to operational normalcy.

Adaptive strategies: Further analysis delves into the continuity of routes over the years. In 2019, only 300 routes persisted into 2020, highlighting disruptions in the majority of existing routes. However, 817 new routes emerged in 2020, showcasing an adaptive strategy to navigate disruptions.

Resilience and rebound: This adaptive strategy proved successful, as evidenced by the continuity of routes in 2021. By utilizing Eqs. 3–5, it is identified that out of the 1582 total routes in 2019, only 300 persisted in 2020, and 703 were found again in 2021 or to put it differently, out of the total of 1618 routes in 2021, 703 is the same routes as those in 2019 and 409 routes persisted from 2020. This indicates that the shipping industry demonstrated resilience and an effective rebound strategy, adapting and recovering its route network efficiently.

4.1.7 Adaptability Analysis

In the analysis spanning from 2019 to 2020, we observe a positive average adaptability index of 0.0759 and a positive average traffic change of 4860.41. This suggests that during the normal state in 2019, the routes demonstrated a moderate level of adaptability, effectively coping with the disruptions experienced in 2020. Conversely, when examining the period from 2020 to 2021, both the average adaptability index (-0.056) and traffic change (-1831.80) exhibit negative values. This indicates that during the disruptive period in 2020, the routes faced challenges in adapting to the subsequent recovery phase in 2021. Further exploration into the 2019 to 2021 time-frame reveals positive values for both indexes (average adaptability index: 0.0172, average traffic change: 1935.71), suggesting that overall, the routes displayed a moderate level of adaptability during the normal state. In order to illustrate further, the visualization is available in Appendix 1 (A, B, C) and its summary in Table 6.

4.2 Deliberate Attack Scenario

We implemented three node removal strategies to simulate disruption: removing the top 20 highest degree ports, the top 20 initial ports, and the top 20 end ports. Regardless of the strategy, the number of similar routes within the same year remained constant, indicating sustained adaptability in service provision. When removing the top 20 highest degree ports, we observed a decrease of 20 routes in 2019 and 19 routes in both 2020 and 2021. Interestingly, the most impacted were unique routes, existing only in a specific year. For instance, removal caused 15 unique routes to disappear in 2019, 18 in 2020, and 13 in 2021. This dynamic can be observed further in Table 3. Further investigation is warranted to explore potential rerouting strategies for network adaptability in future research.

Shifting to the second removal strategy, removing the top 20 initial ports had a milder impact on the total number of routes compared to removing the highest degree ports. However, it significantly disrupted the dynamics of similar routes across different years like 8 routes in 2019 to 2020, 45 routes in 2020 to

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Table 3 Analysis of route dynamics in the GLSN after removing the top 20 highest degree port

	Number of similar and unique routes				Total routes
	2019	2020	2021	Unique	
2019	99	300	698	465	1562
2020	300	22	408	368	1098
2021	698	408	84	409	1599

Table 4 Analysis of route dynamics in the GLSN after removing the top 20 most initial port

	Number of similar and unique routes				Total routes
	2019	2020	2021	Unique	
2019	99	292	669	511	1571
2020	292	22	364	435	1113
2021	669	364	84	491	1608

2021, and 34 routes in 2019 to 2021. At the same time, the removal simultaneously increases the number of unique routes to adapt to changes or disruptions. It is observed that 31 routes in 2019, 49 routes in 2020, and 69 routes in 2021 emerged as an adaptive strategy. These numbers can be calculated when we compare Table 4 with Table 2 as the benchmark. The network’s ability to reconfigure around these disruptions results in the introduction of more unique routes, adapting to the loss of these critical nodes. Therefore, removing the most initial ports do not completely disrupt the network because alternative origins or routing strategies are employed. Further route dynamics after most initial port removal can be observed in Table 4.

Similarly, the third removal strategy, removing the top 20 end ports, also disrupted similar routes across the years, while augmenting the number of unique routes to adapt to the disruption. Nonetheless, the impact of removing the most end ports is deemed lighter compared to removing the most initial ports. It is observed that 4 routes in 2019 to 2020 and 39 routes in 2019 to 2021 experienced a decrease after the removal, while 5 extra routes continued to persist in 2021 following the attack in 2020. The adaptation step is seen when we observed the emergence of 30 and 26 more unique routes in 2019 and 2021.

	Number of similar and unique routes				Total routes
	2019	2020	2021	Unique	
2019	0	− 4	− 39	+ 30	− 13
2020	− 4	0	+ 5	− 3	− 2
2021	− 39	+ 5	0	+ 26	− 8

The number can easily be calculated by comparing Tables 5 and 2. Further route dynamics after most end port removal can be observed in Table 5.

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4.2.1 Adaptability Analysis

Upon analyzing the adaptability following the first removal strategy of the top 20 highest degree ports, it is evident that from 2019 to 2020, there is a slight decrease in the average adaptability index, dropping from 0.0759 to 0.070. Concurrently, there is a slight increase in traffic change, rising from 4860.41 in the baseline scenario to 5799.378. This indicates that while some routes experienced an increase in traffic flow, the overall network’s adaptability diminished. This finding suggests that removing high-degree ports, which often serve as critical connections or central hubs, leads to congestion or inefficiencies in traffic rerouting. While it is a known fact that high-degree node removals negatively impact logistics networks, our analysis quantifies this impact within the specific context of the GLSN. It highlights how the disruption of central hubs can lead to a less adaptable network, where increased traffic on remaining routes may cause bottlenecks and operational challenges. Furthermore, this scenario aligns with the dynamic assessment of the GLSN by Bai et al. [4], supporting their conclusion that high-degree nodes are crucial for maintaining network resilience. Our study adds empirical evidence to this understanding by demonstrating the measurable effects on traffic flow and adaptability when such nodes are removed.

Conversely, in the period from 2020 to 2021, both the average adaptability index (−0.011) and traffic change (2134.37) experience an increase from the baseline scenario (−0.056 and −1831.80, respectively). This suggests that during the disruptive period in 2020, removing the top highest degree ports led to a more efficient redistribution of traffic, albeit still struggling, as indicated by the negative index. However, when considering the period from 2019 to 2021, there is a decrease from the baseline scenario. For further illustration, visualizations are available in Appendix 2 (A, B, C) and its summary in Table 6.

Moving to the second strategy, which involves removing the 20 most initial ports, the results show a slight increase in both the adaptability index and traffic change from 2019 to 2020. This suggests that these initial ports might have experienced congestion during that time, and their removal helped alleviate congestion, improving traffic flow and leading to better adaptation to the disruption. This finding supports the work of Dirzka et al. (2020), stating that during the pandemic, some ports were only partially open [7]. In the period from 2020 to 2021, both the average adaptability index (−0.050) and traffic change (−1463) experienced an increase from the baseline scenario (−0.056 and −1831.80, respectively). This suggests that during the disruptive period in 2020, removing the top initial ports led to a more efficient redistribution of traffic, albeit still struggling, as indicated by the negative index. In 2019 to 2021, the results were almost the same as the baseline, indicating

Table 5 Analysis of route dynamics in the GLSN after removing the top 20 most end port

	Number of similar and unique routes				Total routes
	2019	2020	2021	Unique	
2019	99	296	664	510	1569
2020	296	22	414	383	1115
2021	664	414	84	448	1610

Table 6 Summary of adaptability analysis. It shows average traffic change, average adaptability index, most adaptable route, and most vulnerable route in the normal state (as is), after the highest degree removal, most initial port removal, and most end port removal strategy

Removal strategy	2019 to 2020				2020 to 2021				2019 to 2021			
	ATC	AAI	MA	MV	ATC	AAI	MA	MV	ATC	AAI	MA	MV
As is	4860.41	0.076	699	1155	-1831.8	-0.056	361	11	1935.71	0.017	1227	927
Degree	5799.378	0.070	699	1155	2134.373	-0.010	361	11	114.597	-0.001	1227	975
Initial	4943.85	0.079	699	1155	-1463	-0.050	361	11	1859.768	0.017	1227	972
End	5348.7	0.072	699	1155	427.522	-0.027	361	11	324.11	-0.002	1227	927

ATC average traffic change, AAI average adaptability index, MA most adaptable route, MV most vulnerable route

that the removal did not have a significant impact on adaptability. The visualization can be found in Appendix 3 (A, B, C) and its summary in Table 6.

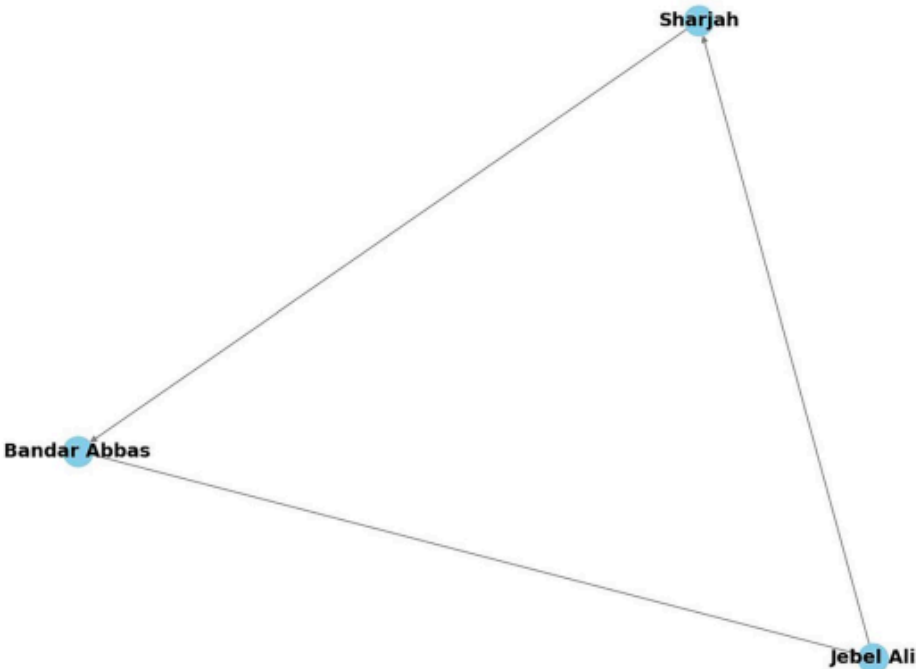
Moving to the third strategy, which involves removing the 20 most end ports, it is surprising that the results show a similar pattern as removing the highest degree ports. In the period from 2019 to 2020, there was a slight decrease in the average adaptability index, dropping to 0.072 compared to the normal state of 0.0759. However, there is a corresponding slight increase in traffic change, rising to 5348.7 from 4860.41 in the normal state. This suggests that although certain routes experience increased traffic flow, the overall network's adaptability decreases. This scenario may indicate that removing most end ports disrupts critical connections or central hubs in the network, leading to congestion or inefficiencies in traffic rerouting, supporting the work of dynamic assessment of GLSN by Bai et al. [4]. Despite the increased traffic, the network may struggle to redistribute it effectively due to the loss of key ports, resulting in decreased adaptability. Conversely, in the period from 2020 to 2021, both the average adaptability index (-0.027) and traffic change (427.522) experience an increase from the baseline scenario (-0.056 and -1831.80 , respectively). This suggests that during the disruptive period in 2020, removing the most end ports led to a more efficient redistribution of traffic, albeit still struggling, as indicated by the negative index. However, when considering the period from 2019 to 2021, there is a decrease from the normal state. For further illustration, visualizations are available in Appendix 4 (A, B, C) and its summary in Table 6.

4.3 Vulnerability Analysis

In terms of adaptability, the GLSN has demonstrated moderate adaptability, particularly in response to significant disruptions during the pandemic in 2020. However, this does not imply that all routes within the GLSN are equally adaptable. Each route has its own adaptability index, and when averaged across the network, it suggests moderate adaptability overall. However, individual routes may exhibit vulnerability to disruptions. In this section, we discuss the results of our analysis. First, focusing on route dynamics, we observe that disruptions did not impact similar routes within the same year, but they did affect similar routes across different years. Removal of the most initial and most end ports resulted in the disruption of such routes, leading to the creation of new unique routes to adapt. On the other hand, removing the highest degree ports primarily disrupted unique routes and had minimal impact on similar routes across different years. Consequently, the highest degree ports are vulnerable to unique routes, while most initial and most end ports are vulnerable to similar routes across different years.

Continuing our analysis, we identify the most adaptable and vulnerable routes. The most adaptable routes are those with the highest adaptability index over the years, while the most vulnerable routes are those with the lowest adaptability index. In 2019 to 2020, Route 699 (Fig. 4) consistently emerges as the most adaptable route across all scenarios, while Route 1155 (Fig. 5) consistently appears as the most vulnerable route. In 2020 to 2021, Route 361 (Fig. 6) is consistently the most adaptable route, while Route 11 (Fig. 7) is consistently the most vulnerable. In the period from 2019 to 2021, Route 1227 (Fig. 8) is the most adaptable across

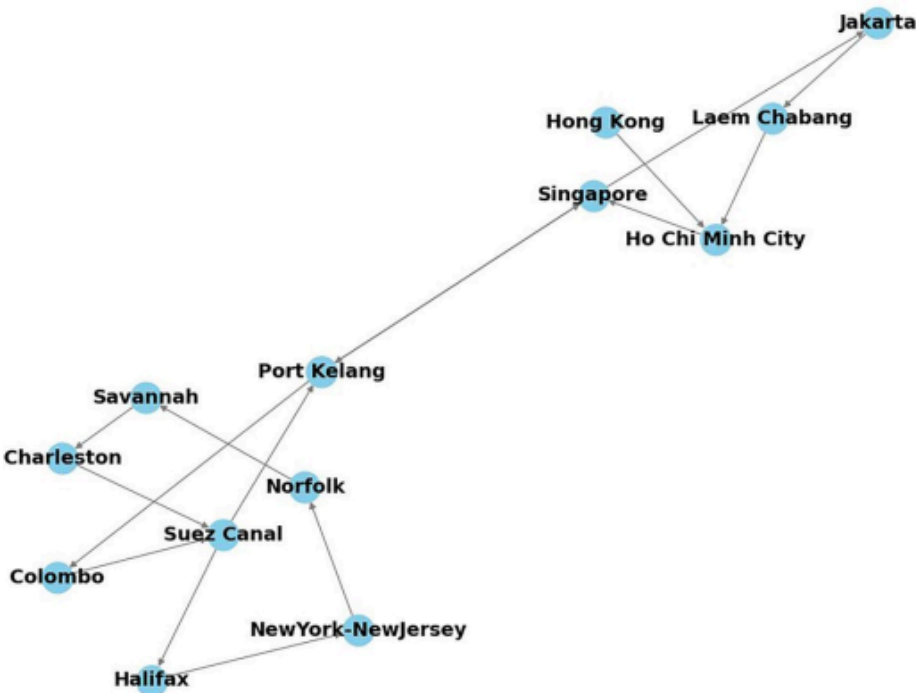
Route 699 on GLSN (Most Adabtable Route 2019 to 2020)



1

Fig. 4 Illustration of Route 699 in network form, the most adaptable route in 2019 to 2020 for all scenarios

Route 1155 on GLSN (Most Vulnerable Route 2019 to 2020)



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Fig. 5 Illustration of Route 1155 in network form, the most vulnerable route in 2019 to 2020 for all scenarios

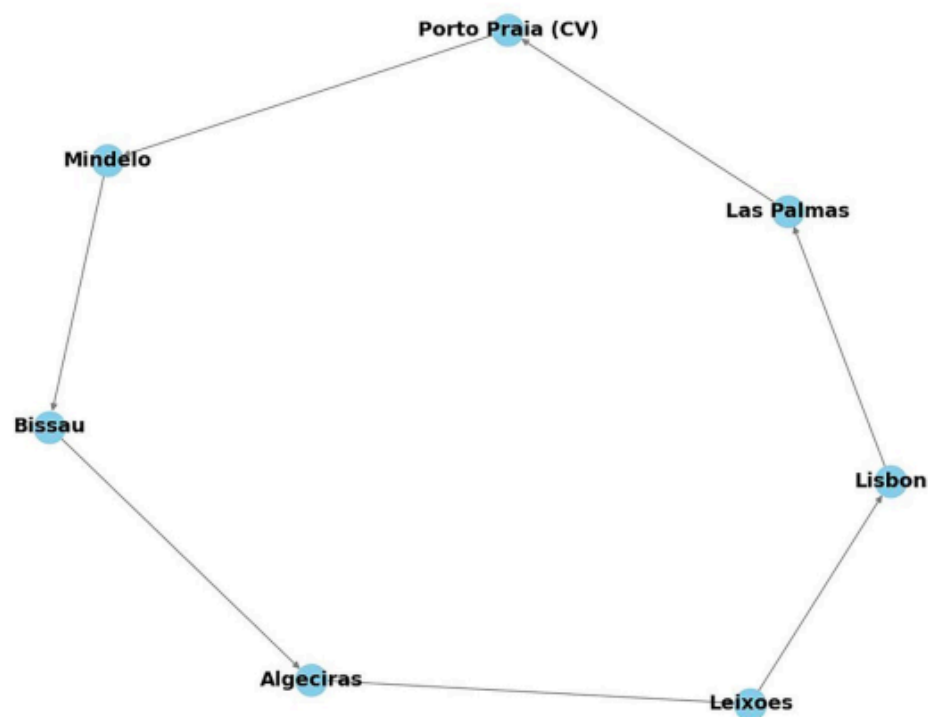
all scenarios, while Route 927 (Fig. 9) emerges as the most vulnerable for the normal state and end port removal scenarios. For the highest degree removal strategy, Route 975 (Fig. 10) is the most vulnerable, while for the most initial port removal strategy, Route 972 (Fig. 11) exhibits the highest vulnerability.

5 Discussion

In terms of the resilience of the global liner shipping network (GLSN), an examination of both its visual representation and structure unequivocally demonstrates its proven resilience, even in the face of significant disruptions like the pandemic in 2020. All metrics employed in this study consistently indicate a substantial decline in GLSN's performance in 2020, followed by a resilient rebound in 2021. This concept of resilience was also highlighted by (1) referencing some of the previous work about adaptability, robustness, and recoverability [2, 3, 16]. The network is considered resilient when, after a disruption, it successfully returns to a normal state.

In terms of adaptability, while the overall Global Liner Shipping Network (GLSN), as represented by the average adaptability index, demonstrates moderate adaptability both in normal states and during recovery periods, it is noteworthy that the index values are relatively small. This suggests that there is room for improvement. Further investigation using optimization approaches for adaptability can be implemented to delve deeper into the route dynamics within the GLSN.

Route 361 on GLSN (Most Adaptable Route 2020 to 2021)



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Fig. 6 Illustration of Route 361 in network form, the most adaptable route in 2020 to 2021 for all scenarios

Route 11 on GLSN (Most Vulnerable Route 2020 to 2021)

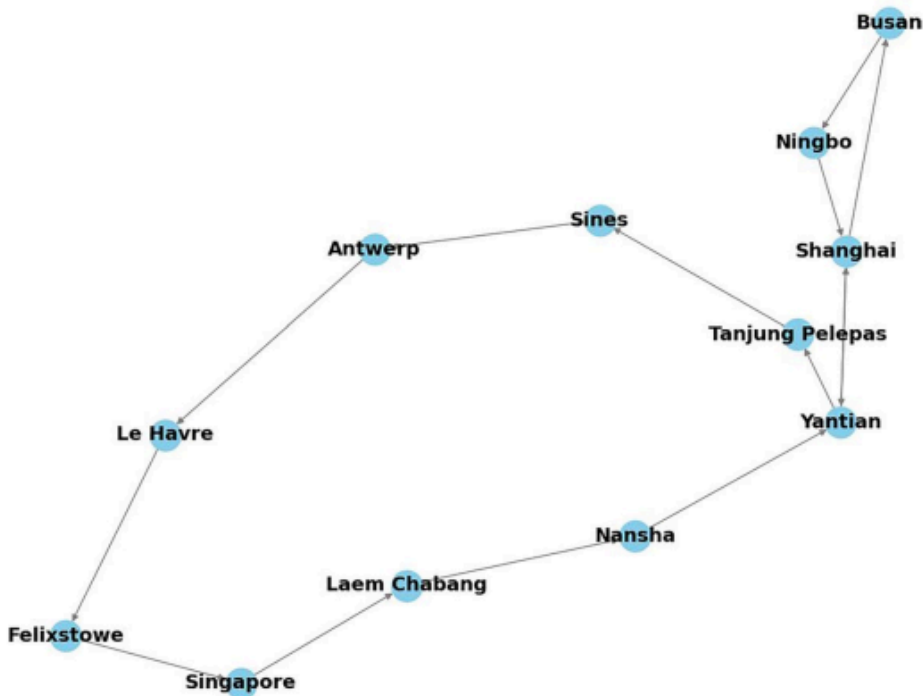


Fig. 7 Illustration of Route 11 in network form, the most vulnerable route in 2020 to 2021 for all scenarios

Route 1227 on GLSN (Most Adaptable Route 2019 to 2021)

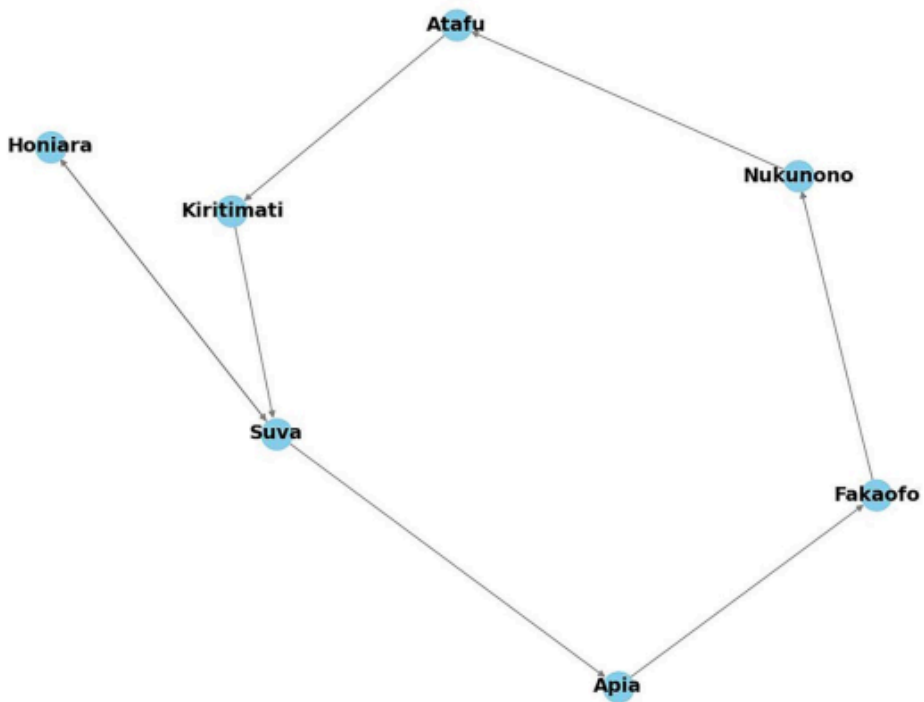


Fig. 8 Illustration of Route 1227 in network form, the most adaptable route in 2019 to 2021 for all scenarios

Route 927 on GLSN (Most Vulnerable Route 2019 to 2021 for normal state & most end port removal)

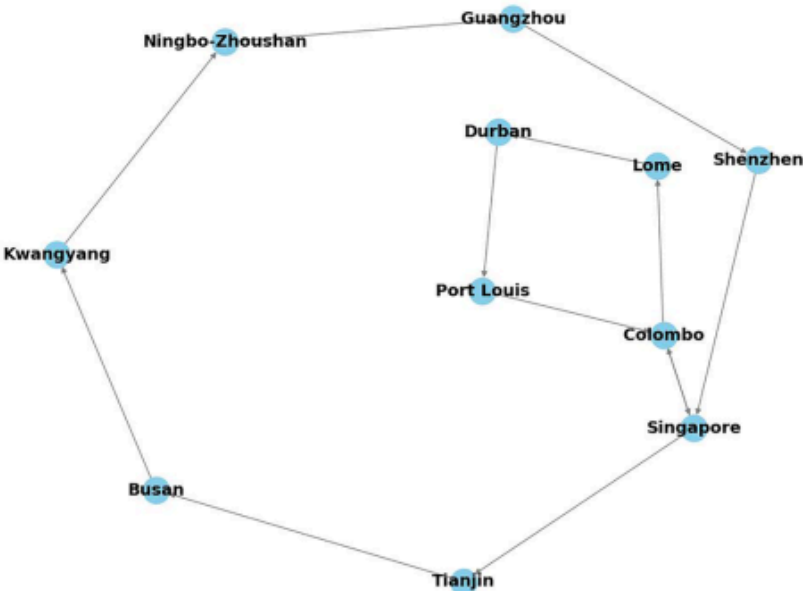


Fig. 9 Illustration of Route 927 in network form, the most vulnerable route in 2019 to 2021 for normal state and most end port removal

Route 975 on GLSN (Most Vulnerable Route 2019 to 2021 for Highest Degree removal)

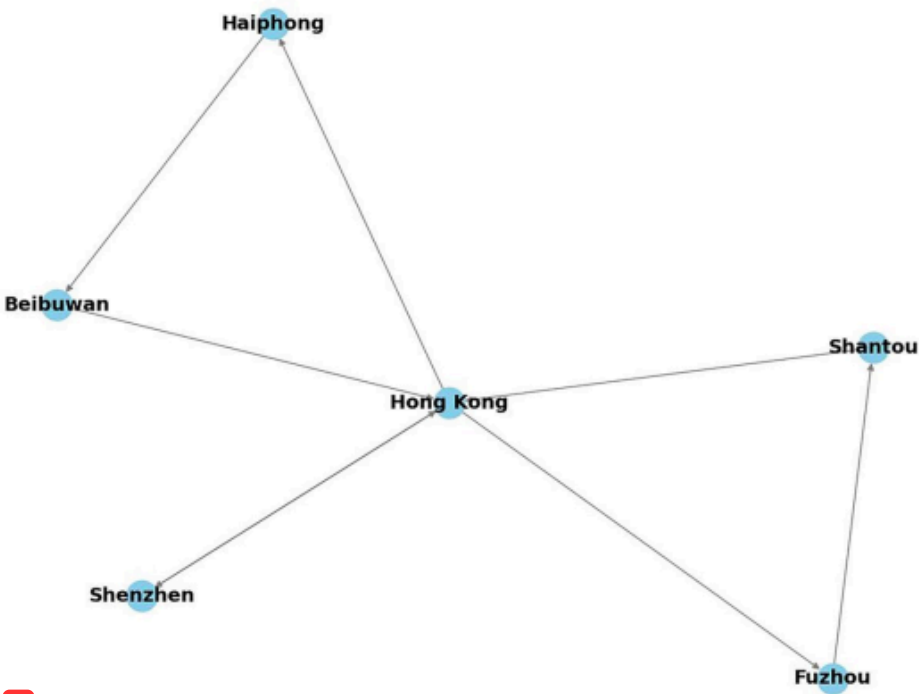


Fig. 10 Illustration of Route 975 in network form, the most vulnerable route in 2019 to 2021 for the highest degree removal

Regarding route vulnerability, it is crucial to note that the most vulnerable route changes each year, particularly depending on which port is being targeted. This study sheds light on how to pinpoint the exact routes vulnerable to specific attacks, facilitating discussions on mitigation actions to prevent such vulnerabilities from being exploited.

6 Conclusion

The pandemic has wrought unprecedented impacts on various facets of life, notably affecting the supply chain, with a particular emphasis on the global liner shipping network (GLSN). The disruptive force of the pandemic has served as a litmus test for the resilience of the GLSN. The network has demonstrated its resilience by effectively returning to a normal operational state in the year following the disruption. All metrics employed in this study consistently affirm this outcome, indicating a substantial disturbance in the GLSN in 2020, followed by a successful restoration to a normal state in 2021. Consequently, the network is deemed resilient.

Through network analysis in the “as is” scenario, we have determined that the Global Liner Shipping Network (GLSN) exhibits moderate adaptability overall, particularly when operating under normal conditions. However, during disruptive periods such as 2020, the adaptability index showed instability, indicating potential challenges in responding to major disruptions. In the deliberate attack scenario, most strategies resulted in moderate adaptability, except during the disruption period in

Route 972 on GLSN (Most Vulnerable Route 2019 to 2021 for Most Initial Port removal)

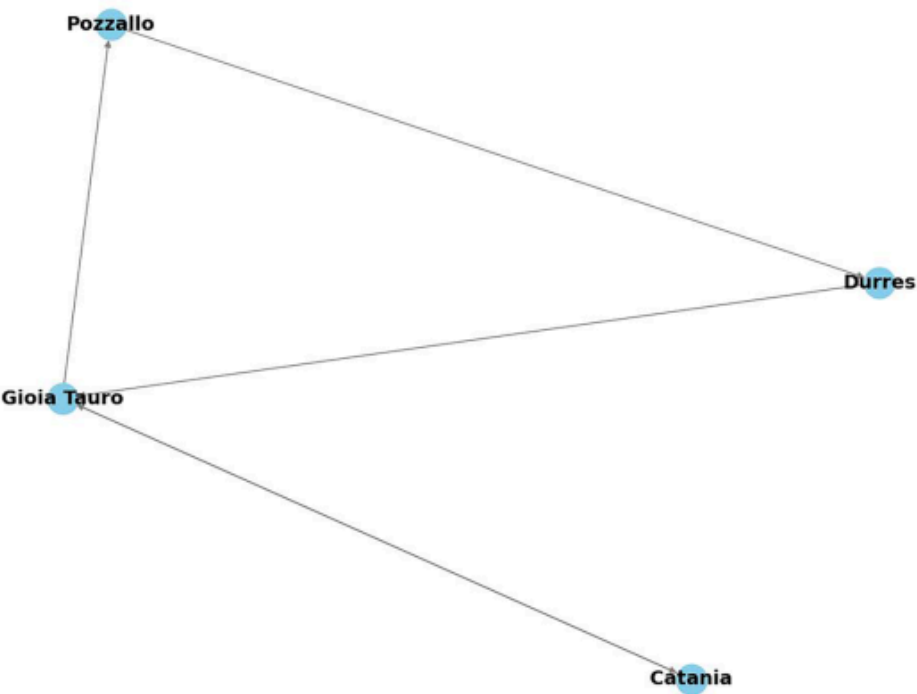


Fig. 11 Illustration of Route 972 in network form, the most vulnerable route in 2019 to 2021 for most initial port removal

2020 where all strategies showed negative adaptability. Notably, the highest degree removal strategy disrupted unique routes in specific years, while attacks on the most initial and end ports disrupted similar routes across multiple years.

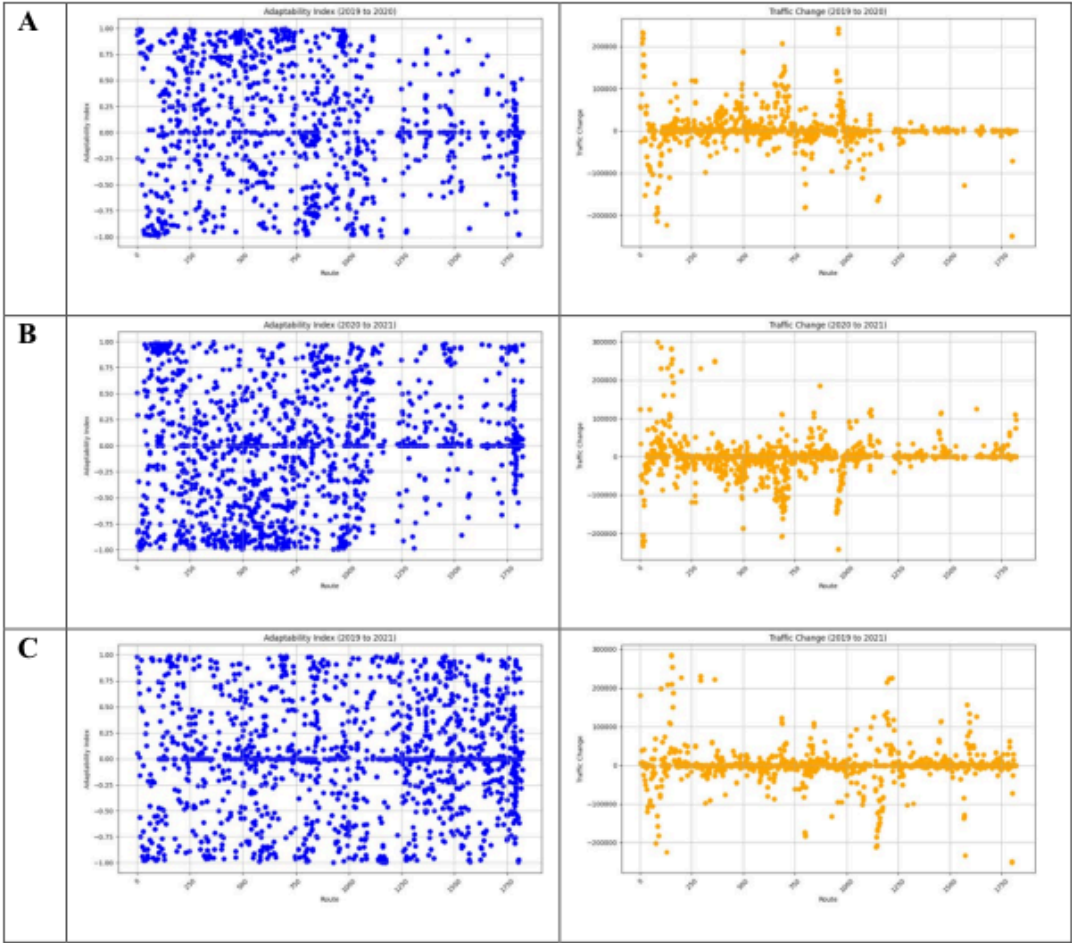
Furthermore, our study highlights the dynamic nature of route vulnerability, with the most vulnerable routes changing over time and depending on the type of port targeted. This insight underscores the importance of further research into mitigating disruptions to ensure the resilience and adaptability of the GLSN in the face of evolving challenges.

Appendix 1

Appendix 1 shows the adaptability index and traffic change graph in the “as is” scenario.

The adaptability index graph showcases the adaptability of each route, represented by blue dots. Closer proximity to 1 indicates higher adaptability, while proximity to negative 1 signifies greater vulnerability.

Meanwhile, the traffic change graph illustrates traffic fluctuations on each route, with yellow dots. Positive values indicate increasing traffic, while negative values denote decreasing traffic.

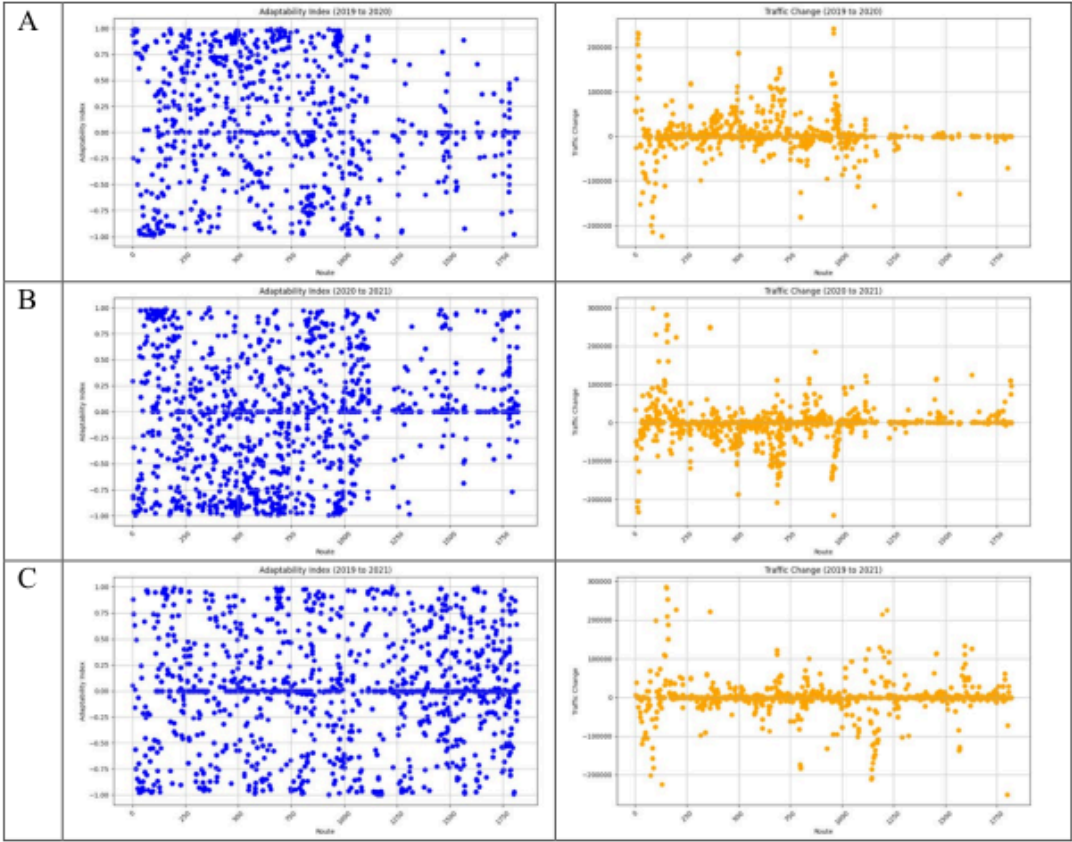


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Appendix 2

Appendix 2 shows the adaptability index and traffic change graph in a deliberate attack scenario for the highest-degree port removal strategy.

The adaptability index graph showcases the adaptability of each route after removal, represented by blue dots. Closer proximity to 1 indicates higher adaptability, while proximity to negative 1 signifies greater vulnerability.

Meanwhile, the traffic change graph illustrates traffic fluctuations on each route after removal, with yellow dots. Positive values indicate increasing traffic, while negative values denote decreasing traffic.



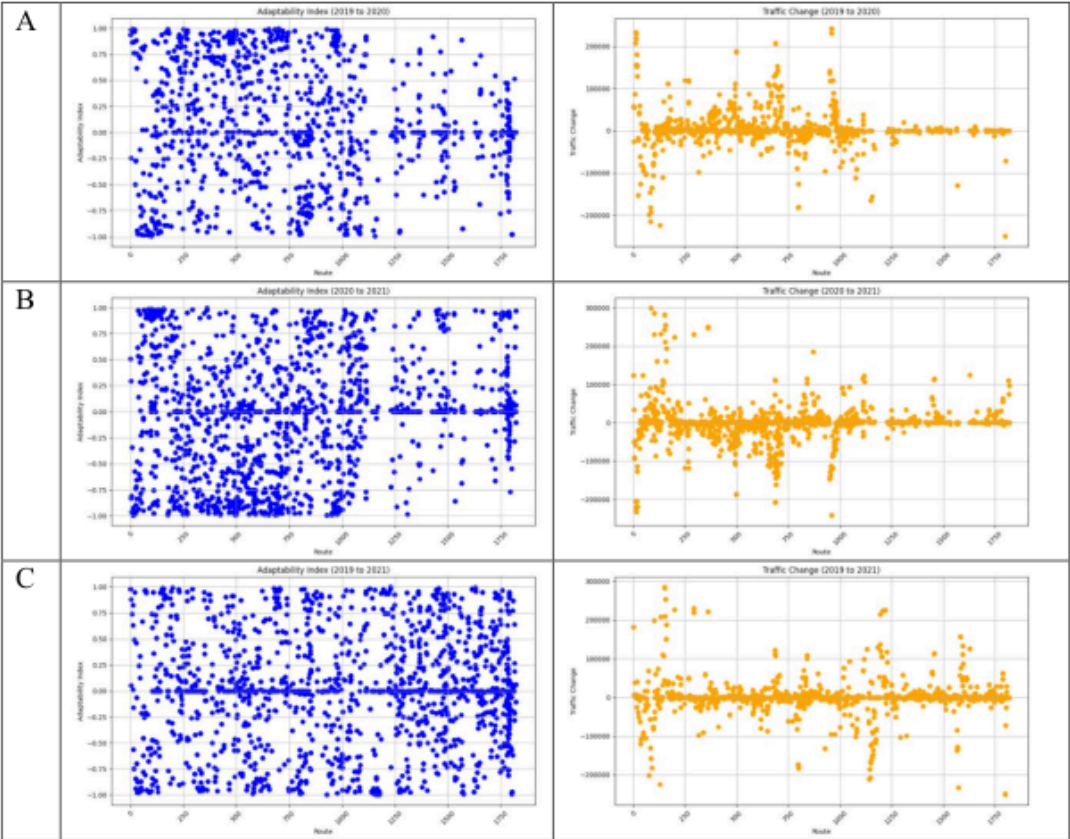
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Appendix 3

Appendix 3 shows the adaptability index and traffic change graph in the deliberate attack scenario for most initial port removal strategy.

The adaptability index graph showcases the adaptability of each route after removal, represented by blue dots. Closer proximity to 1 indicates higher adaptability, while proximity to negative 1 signifies greater vulnerability.

Meanwhile, the traffic change graph illustrates traffic fluctuations on each route after removal, with yellow dots. Positive values indicate increasing traffic, while negative values denote decreasing traffic.

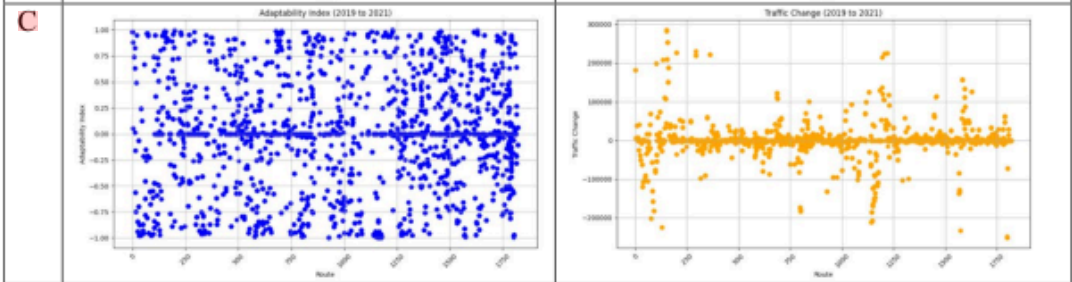
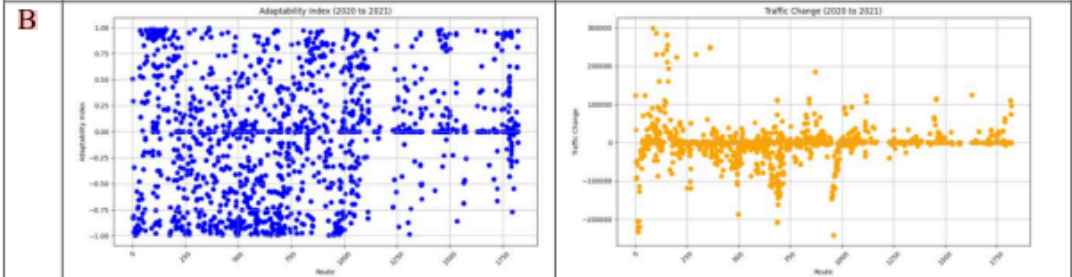
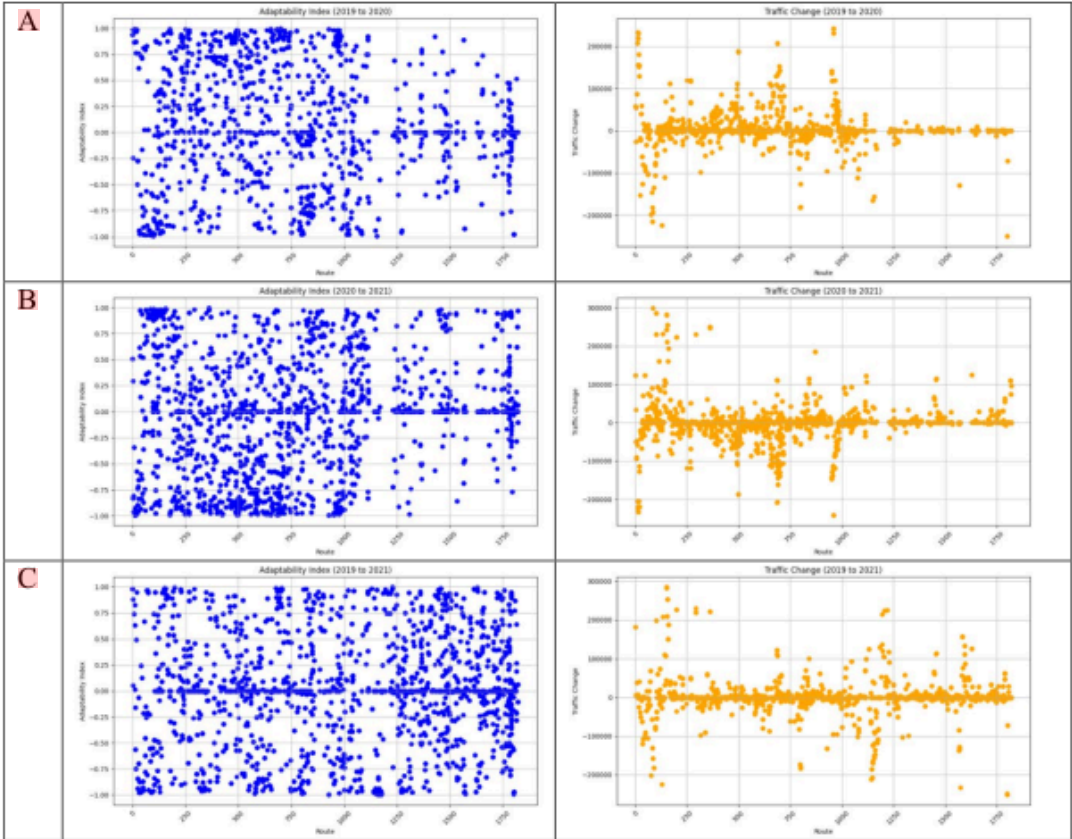


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Appendix 4

Appendix 4 shows the adaptability index and traffic change graph in the deliberate attack scenario for most end port removal strategy.

The adaptability index graph showcases the adaptability of each route after removal, represented by blue dots. Closer proximity to 1 indicates higher adaptability, while proximity to negative 1 signifies greater vulnerability.

Meanwhile, the traffic change graph illustrates traffic fluctuations on each route after removal, with yellow dots. Positive values indicate increasing traffic, while negative values denote decreasing traffic.



Author Contribution W.T.A wrote the main manuscript text, Q.P prepared the data and H.X. supervised and validated the whole works. All authors reviewed the manuscript.

Data Availability No datasets were generated or analysed during the current study.

Declarations

Competing Interests The authors declare no competing interests.

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